

Etienne Mueller

(address hidden in web version)

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PROFILE

Innovative AI and computational neuroscience professional with a history of developing advanced algorithms, machine learning systems, and neuromorphic computing solutions across academia, startups, and leading global organizations. Proven track record of designing efficient AI models/tools that reduce processing times, enhance accuracy, and lower costs. Skilled in conducting AI research, designing/deploying cutting-edge tools, and using high-powered computing to analyze complex data sets.

PROFESSIONAL EXPERIENCE

Assistant Professor AI & Data Science

2025 – Present

Carnegie Mellon University (CMKL), Thailand

- Investigating hierarchical spiking neural networks for cognitive neuromorphic computing, focusing on bio-inspired multi-timescale architectures for compositional reasoning with planned validation on various neuromorphic hardware
- Developing the curriculum for *Neural Networks and Deep Learning*, *LLMs and Agentic AI*, and *Big Data Computing*

Postdoctoral Researcher AI

2023 – 2025

University of Melbourne, Australia

- Developed neural network growth algorithms to create biologically-inspired memory cells for more efficient recurrent neural networks, implementing ML simulations of brain imaging data across different developmental stages using JAX
- Ran deep convolutional neural networks on a Slurm-based HPC with up to 4xH100 GPUs per node for automated segmentation of synchrotron brain imaging data, reducing the need for manual annotation by a factor of 10

AI Engineer

2022 – 2023

Flowers Software, Germany

- Established the AI research department at a seed-financed startup, deploying deep learning infrastructure from scratch
- Developed a TensorFlow-based information extraction workflow on AWS to identify recurring positions on invoices that complies with EU data privacy law, saving the company over €5,000/month by eliminating third-party API costs

AI Researcher

2018 – 2022

Infineon Technologies, Germany

- Research in neuromorphic computing and spiking neural networks, leading to 11 first- and second-author publications
- Developed a TensorFlow-based toolbox for converting conventional to spiking neural networks, which was subsequently used in a research project to reduce the simulation time of hardware components in neuromorphic systems by half

Component Manager

2017 – 2018

BMW AG

- Technical supervision of cooling water pumps for electric and combustion vehicles for BMW, Mini and Rolls Royce
- Requirement engineering and long-term testing of different models in cooperation with Bosch and Continental

Formula Student Member @ e-gnition Hamburg e.V.

2012 – 2017

Developer Driverless Actuator Technology (2016 – 2017)

- 1st Place Formula Student Driverless: Autonomous Design
- 3rd Place Formula Student Driverless: Overall

Division Manager Business Plan (2014 – 2016)

- Special Award for educational video "How to Business Plan" at Formula Student Hungary

President & Team Captain (2013 – 2014)

- Special Award for Ecological Design by Magna Steyr

Division Manager Aerodynamics (2012 – 2013)

Co-founder & CEO

2015 – 2016

Slive Technologies, Germany

- Developed smart wearable devices and location-based algorithms for hands-free data use in industrial environment
- Secured the Nissen Foundation Start-Up Grant (€3,000) to support early-stage product development and business growth

EDUCATION

Nengo Summer School (Neuromorphic AI Systems) <i>University of Waterloo, Canada</i>	2023
Ph.D. in Computer Science (Artificial Intelligence) <i>Technical University Munich, Germany</i>	2023
M.Sc. in Product Development (Autonomous Driving) <i>Technical University Hamburg, Germany</i>	2017
Semester Abroad <i>Institut Catholique d'Arts et Métiers Nantes, France</i>	2016
B.Sc. in Mechanical Engineering (Robotics) <i>Technical University Hamburg, Germany</i>	2014

SKILLS

Programming	Python, TensorFlow, JAX, Bash, Swift, Xcode, C++, SQL
Languages	German (native), French (native), English (fluent), Spanish (basic), Chinese (basic)
Interests	Music (Piano, Ukulele), Fitness, Health

OPEN SOURCE PROJECTS

High-Performance Zebrafish (HPZ)

- A Python and bash toolkit designed to automate recurring brain imaging data tasks on a Slurm-based HPC setup
- End-to-end pipeline that consolidates multiple manual steps for loading, preprocessing, and detecting neurons and spikes in microscopy data into a single automated process, reducing manual intervention and error by a factor of five
- Automated setup for new users to easily work with zebrafish brain imaging data, improving onboarding efficiency

Convert2SNN

- A TensorFlow-based library that converts conventionally trained neural networks with continuous activation functions to spiking neural networks, with minimal to no performance loss, to estimate energy consumption in neuromorphic systems
- Supports key spike encoding techniques, including rate, population, and temporal coding, with the ability to estimate spike counts for efficiency evaluation, reducing the need for extensive hardware simulations during development

Razer Blade for Data Science

- Collection of instructions, settings, and bash scripts for setting up Nvidia-based gaming notebooks to work with Linux and CUDA

SELECTED PUBLICATIONS

E. Mueller, W. Qin, "Reverse Engineering Neural Connectivity: Mapping Neural Activity Data to Artificial Neural Networks for Synaptic Strength Analysis," in *8th International Conference on Information Technology (InCIT)*, Chonburi, Thailand and Kanazawa, Japan, 2024, pp. 451-454.

E. Mueller, S. Klimaschka, D. Auge, A. Knoll, "Neural Oscillations for Energy-Efficient Hardware Implementation of Sparsely Activated Deep Spiking Neural Networks," in *Association for the Advancement of Artificial Intelligence (AAAI) Practical DL*, Online (Vancouver, Canada), 2022, pp. 1-7.

E. Mueller, V. Studenyak, D. Auge, A. Knoll, "Spiking Transformer Networks: A Rate Coded Approach for Processing Sequential Data," in *7th Int. Conference on Systems and Informatics (ICSAI)*, Online (Jiaxing, China), 2021, pp. 1-5.

E. Mueller, J. Hansjakob, D. Auge, A. Knoll, "Minimizing Inference Time: Optimization Methods for Converted Deep Spiking Neural Networks," in *International Joint Conference on Neural Networks (IJCNN)*, Online (Shenzhen, China), 2021, pp. 1-8.

D. Auge, J. Hille, **E. Mueller**, A. Knoll, "A Survey of Encoding Techniques for Signal Processing in Spiking Neural Networks," *Neural Processing Letters*, vol. 53, issue 6, pp. 4693-4710, Dec 2021.

See [Google Scholar profile](#) for a full list.